

FYKE SIMON V. TONEL

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Education

STI College Caloocan

Caloocan City, Metro Manila

Bachelor of Science in Computer Science. GWA: 1.27

Expected 2026

- President's Lister 2022-Present, Alumni Scholar (50% merit)
- Best Thesis & Best Research Project, ICTE Symposium (campus-wide CS)

Experience

Back-End Developer (Intern) – Inspire Holdings Inc.

January 2026 – February 2026

- Built and deployed a NestJS payment backend on AWS (Lambda, API Gateway, RDS), integrating secure UnionBank APIs.
- Reduced **p95** latency from **420ms to 180ms (-57%)** and CPU usage by **35%** using Redis caching and token-bucket rate limiting
- Enforced **API security via HMAC signing** and request validation.
- Delivered **CI/CD pipelines** and **40+ unit tests** using GitHub Actions and Jest.

Software Developer (Intern) – Socia I.T. Solutions

February 2026 – March 2026

- Migrated a web-based streaming platform to **React Native (Expo)**, consolidating iOS and Android into a unified codebase.
- Implemented **WebSocket-driven real-time features** for live streaming and interactive rewards.
- Strengthened runtime reliability by enforcing **TypeScript strict mode** and structured validation.

Projects

Flick Player: Music Player with Custom UAC 2.0

January 2026 – Present

Flutter, Dart, Rust, and Kotlin

- Engineered **four Rust-based audio engines** enabling direct USB output and bypassing Android audio pipeline constraints.
- Architected a hybrid differential music library scanner using **Android MediaStore**, **Rust parallel traversal**, and **fingerprint caching**, reducing scan times from **~11s to 85–328ms (up to 34× faster)** across **1,000+ track SD-card libraries**.
- Optimized music library processing (scanning, metadata extraction, indexing), improving **performance on large libraries**.
- Open-source project with **600+ installs/downloads** across **GitHub and Play Store distributions**.

br41ndmg: High-Fidelity Audio Resampling Library

January 2026 – Present

Rust (DSP and Audio Processing)

- Implemented **polyphase sinc-based resampling** for high-precision sample rate conversion.
- Validated DSP performance using impulse, sinc, and sweep testing with **Criterion benchmarks**.
- Designed configurable **FIR filters achieving >100 dB stopband attenuation and <0.1 dB passband ripple**.

PASADA: Ride-Hailing & Fleet Management Ecosystem

December 2024 – November 2025

Flutter, TypeScript (Bun/Deno), React, Supabase, and Google Cloud Platforms

- Architected a distributed system spanning **4 applications** (Passenger, Driver, Admin, Web) with shared authentication and RBAC.
- Developed a data pipeline for **7-day demand forecasting** and analytics with integrated AI-powered summaries to surface actionable insights from real-time transport data.

Technical Skills

Languages: Rust, Dart, TypeScript, Kotlin, Java, SQL

Backend: Node.js, NestJS, PostgreSQL, Redis, Supabase

Frontend/Mobile: Flutter, React Native, React, Next.js

Cloud/DevOps: AWS, GCP, Docker, Cloudflare, CI/CD

Systems: WebSockets, API Design, Rate Limiting, Real-Time Systems